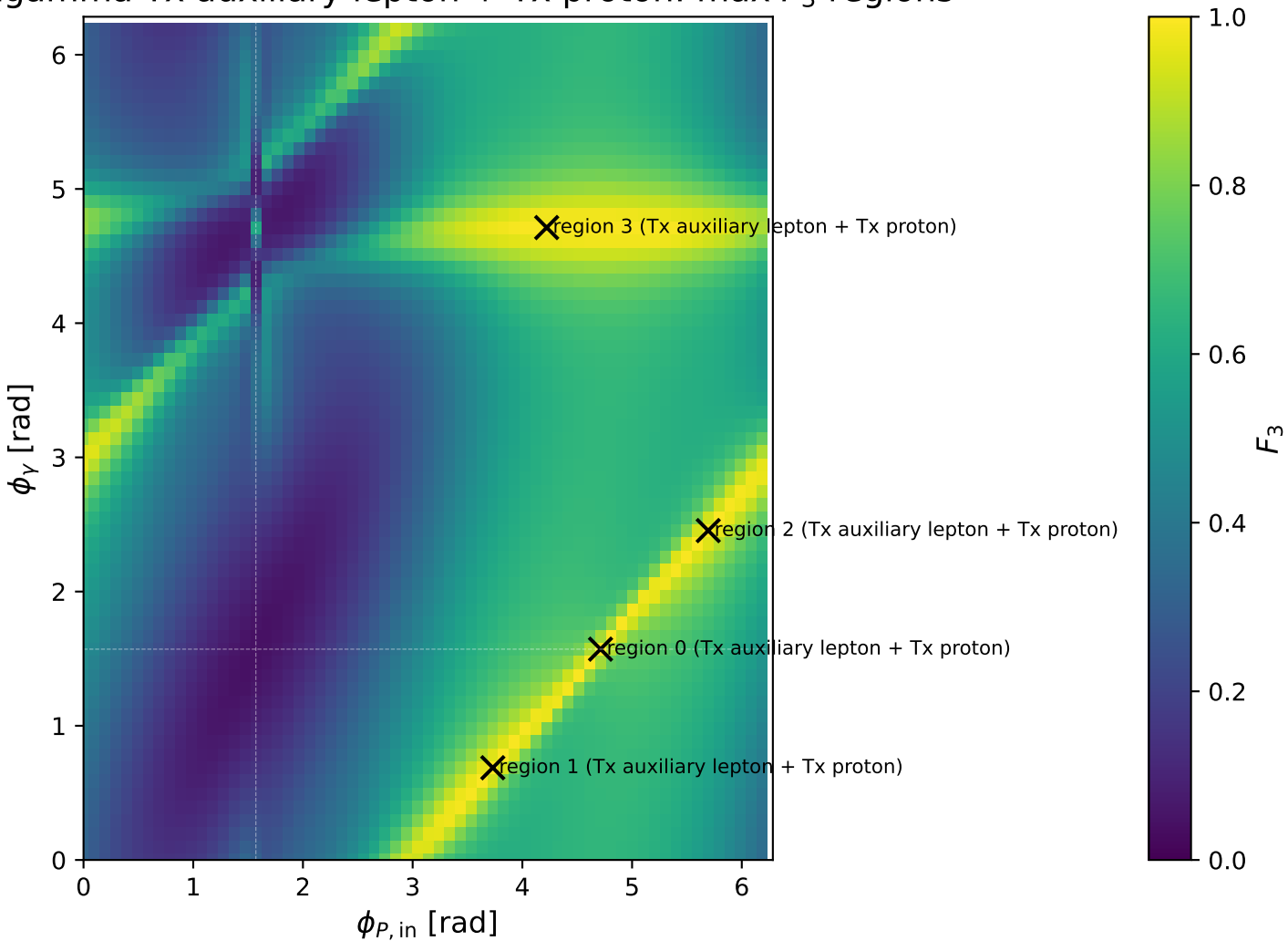
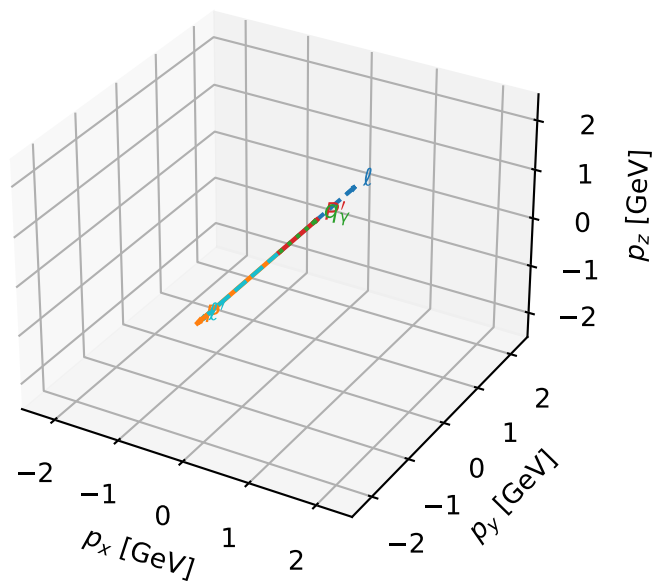


gamma Tx auxiliary lepton + Tx proton: max F_3 regions



Tx auxiliary lepton + Tx proton characteristic max F_3 configuration

3D momenta

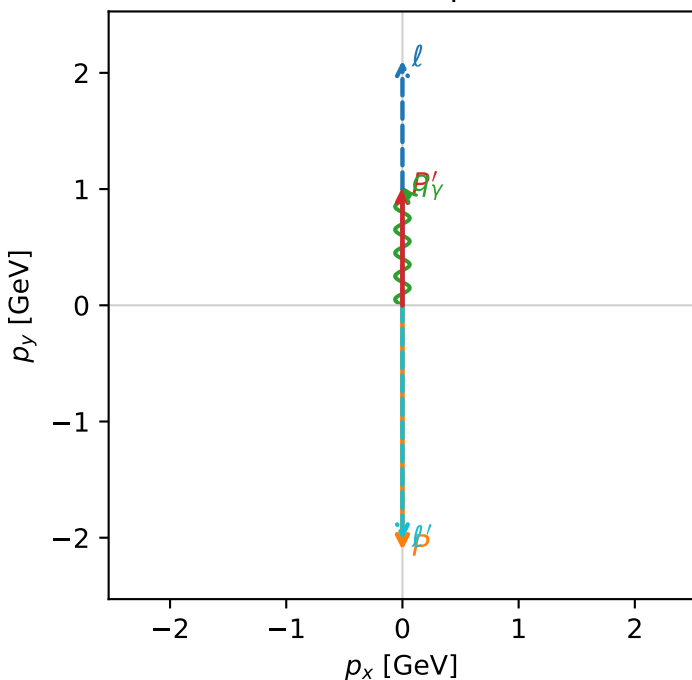


f3_mid_egamma_txtx_region_0 F_3=0.996287
 region: mid_Egamma_TxTx_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,\ell}\rangle \otimes |\bar{T}_{x,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.01927$
 $\theta_{in}=1.57079$, $\phi_\ell=1.5708$, $\phi_p=4.71239$
 $E_\gamma=1$, $\phi_\gamma=1.5708$
 $Q^2=17.0194$, $x_B=0.958772$, $t=-8.92478$, $W^2=1.61168$, $y=0.904017$
 $\theta(\ell', \gamma)=180$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= 0.0000$ $p_y= 2.1069$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -0.0000$ $p_y= -2.1069$ $p_z= 0.0000$
 ℓ' $E= 2.2533$ $p_x= -0.0000$ $p_y= -2.0193$ $p_z= 0.0000$
 P' $E= 1.3852$ $p_x= 0.0000$ $p_y= 1.0193$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.0000$ $p_y= 1.0000$ $p_z= 0.0000$

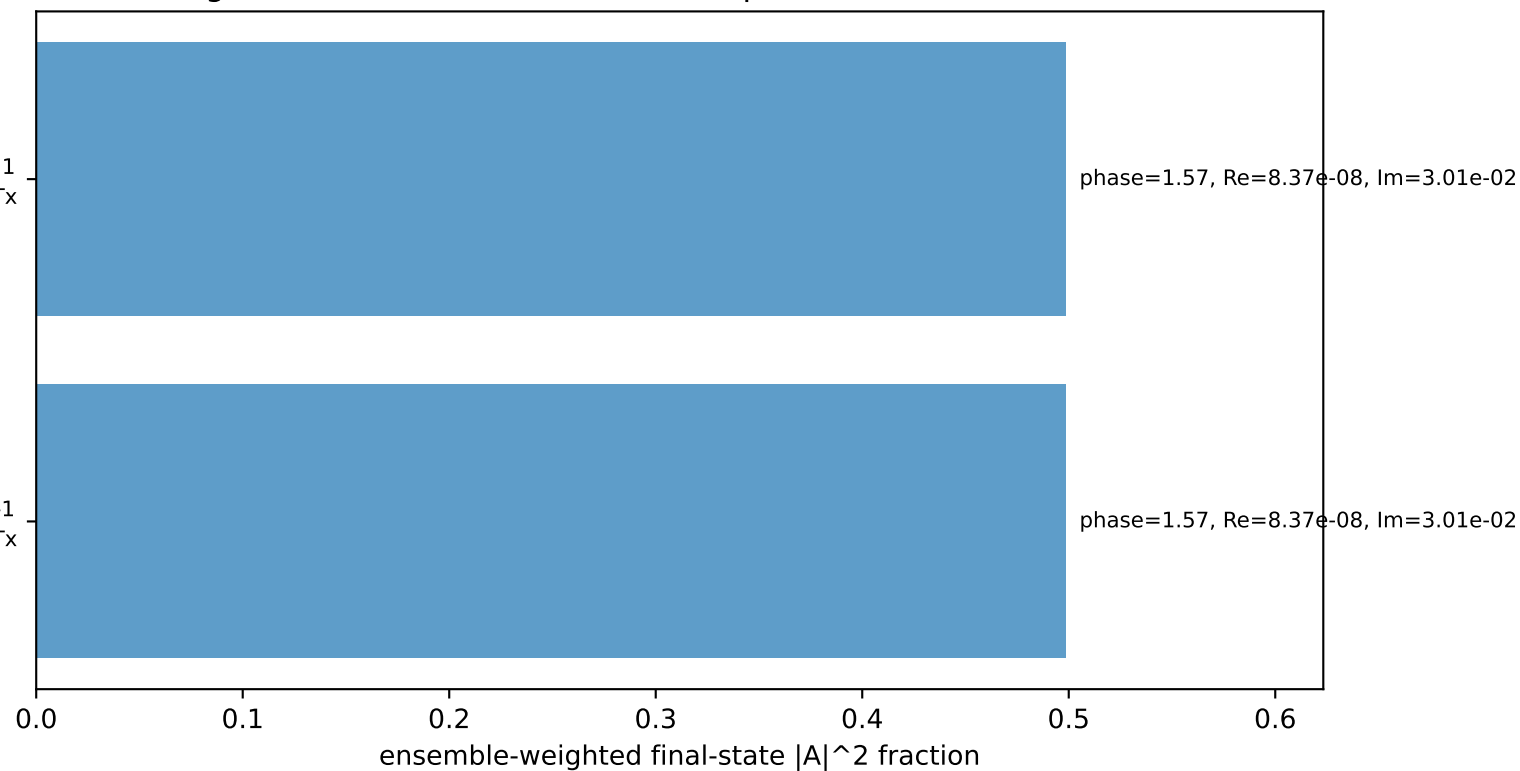
Transverse plane



$h_e=-1, h_p=-1, h_\gamma=+1$
 auxiliary lepton Tx, proton Tx

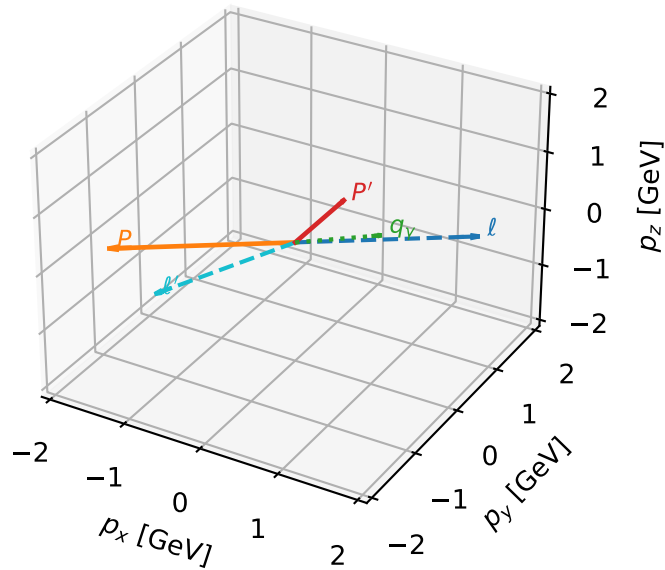
$h_e=+1, h_p=+1, h_\gamma=-1$
 auxiliary lepton Tx, proton Tx

Leading initial-ensemble/final-state components (N=2, retained=99.7%)



Tx auxiliary lepton + Tx proton characteristic max F_3 configuration

3D momenta

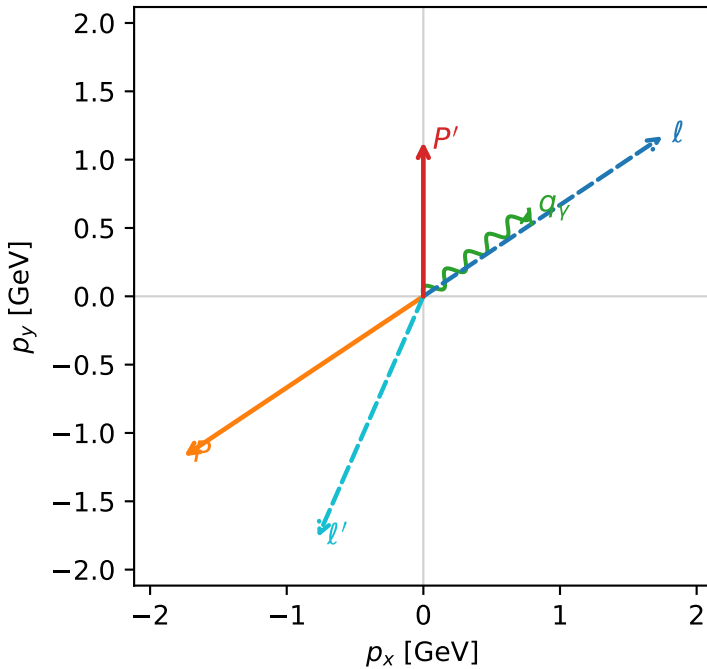


f3_mid_egamma_txtx_region_1 F_3=0.994673
 region: mid_Egamma TxTx_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,\ell}\rangle \otimes |T_{x,p}\rangle$

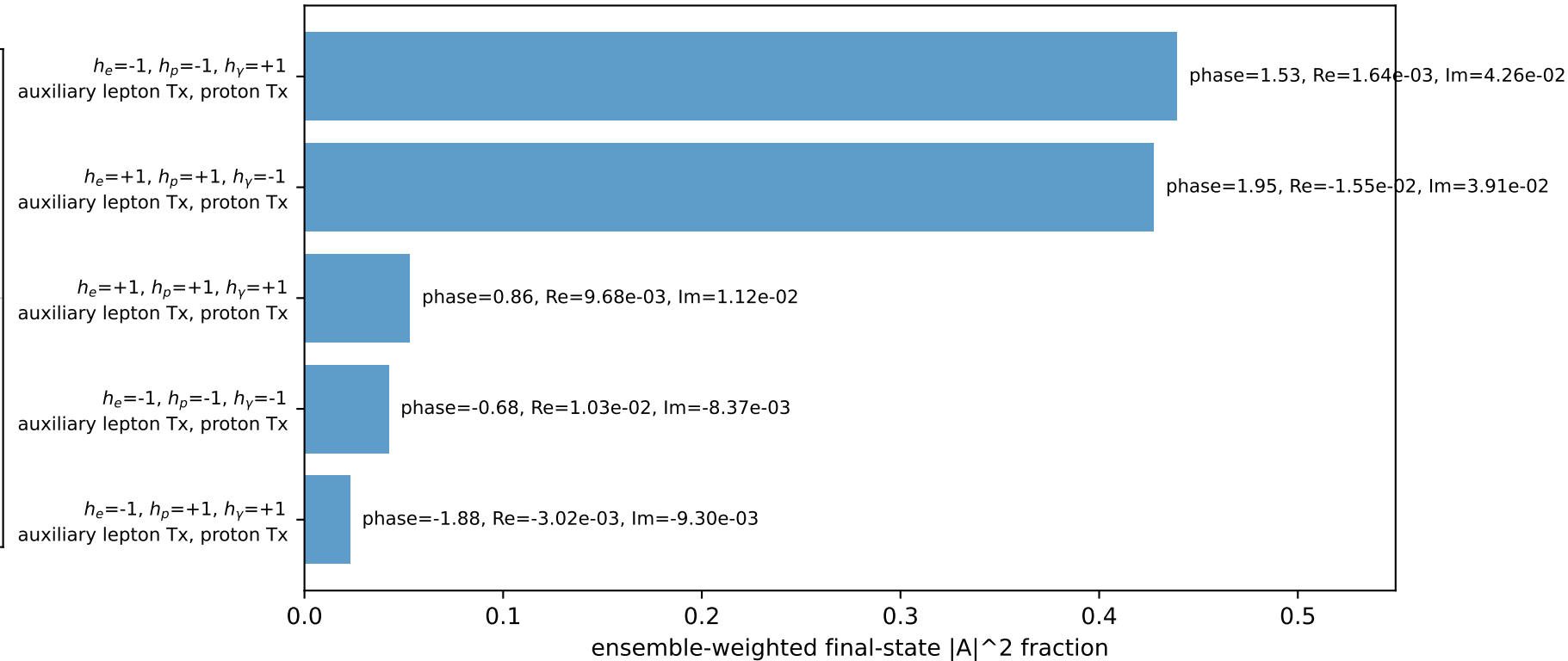
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.12971$
 $\theta_{in}=1.57079$, $\phi_\ell=0.589049$, $\phi_p=3.73064$
 $E_\gamma=1$, $\phi_\gamma=0.687223$
 $Q^2=14.9609$, $x_B=0.908689$, $t=-7.65805$, $W^2=2.38321$, $y=0.838473$
 $\theta(\ell', \gamma)=153.038$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= 1.7519$ $p_y= 1.1706$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -1.7519$ $p_y= -1.1706$ $p_z= 0.0000$
 ℓ' $E= 2.1702$ $p_x= -0.7730$ $p_y= -1.7641$ $p_z= 0.0000$
 P' $E= 1.4684$ $p_x= 0.0000$ $p_y= 1.1297$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.7730$ $p_y= 0.6344$ $p_z= 0.0000$

Transverse plane

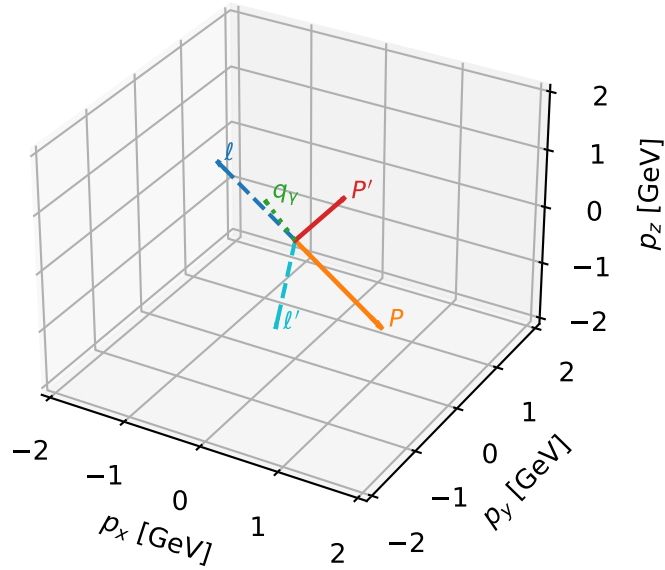


Leading initial-ensemble/final-state components (N=5, retained=98.6%)



Tx auxiliary lepton + Tx proton characteristic max F_3 configuration

3D momenta

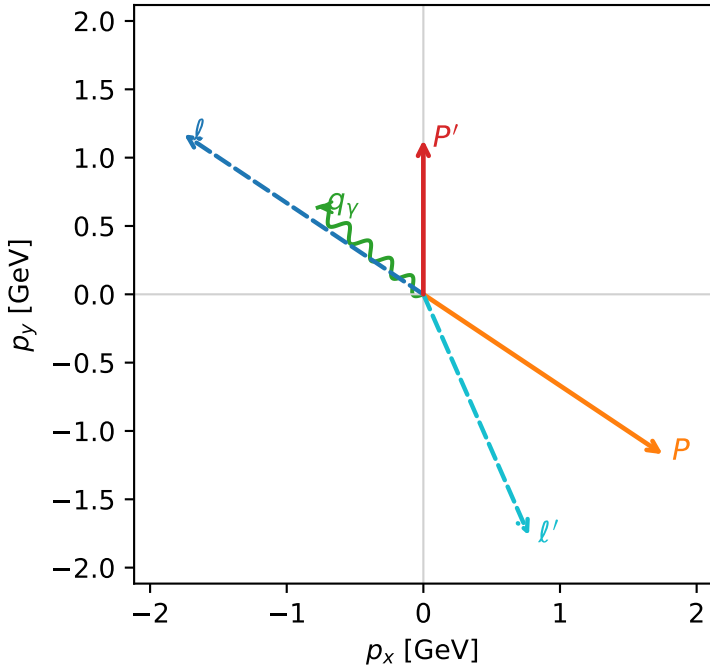


f3_mid_egamma_txtx_region_2 F_3=0.994673
 region: mid_Egamma TxTx_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,\ell}\rangle \otimes |T_{x,p}\rangle$

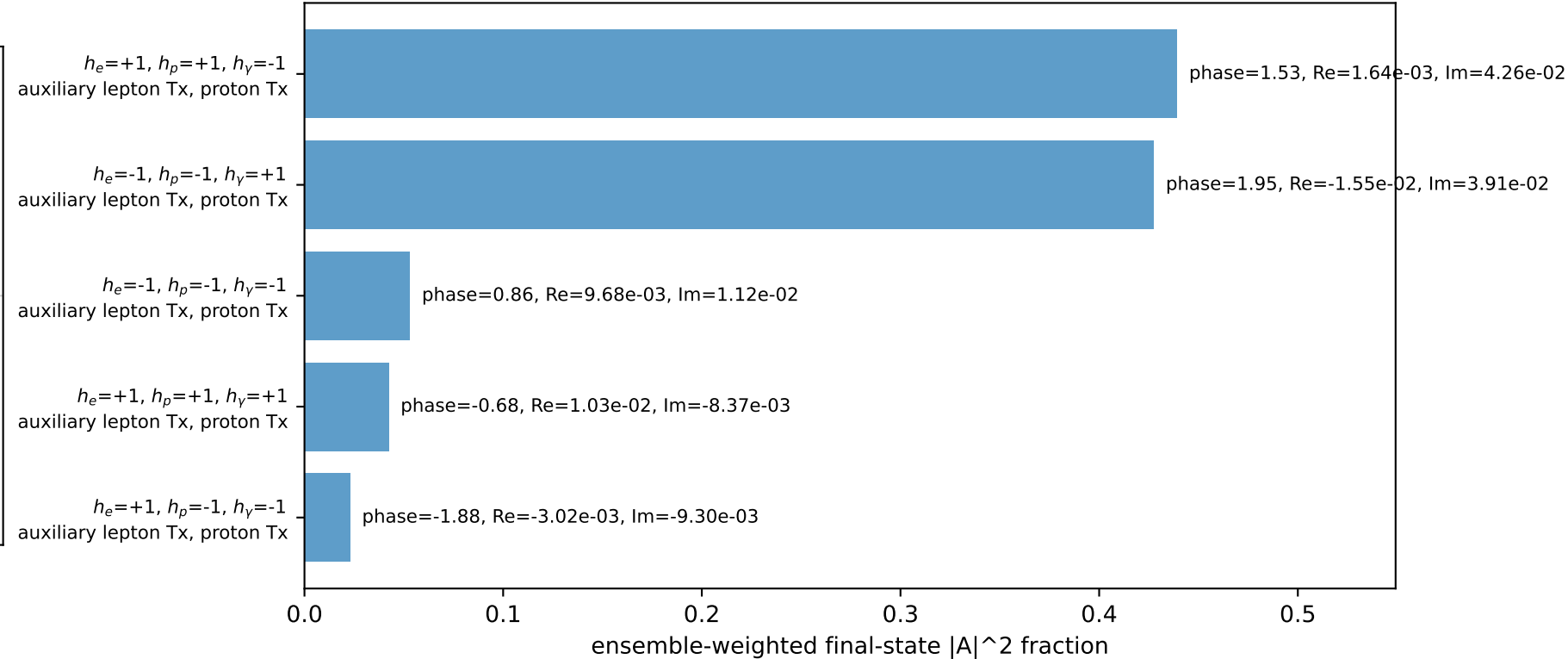
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.12971$
 $\theta_{in}=1.57079$, $\phi_\ell=2.55254$, $\phi_p=5.69414$
 $E_\gamma=1$, $\phi_\gamma=2.45437$
 $Q^2=14.9609$, $x_B=0.908689$, $t=-7.65805$, $W^2=2.38321$, $y=0.838473$
 $\theta(\ell', \gamma)=153.038$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= -1.7519$ $p_y= 1.1706$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 1.7519$ $p_y= -1.1706$ $p_z= 0.0000$
 ℓ' $E= 2.1702$ $p_x= 0.7730$ $p_y= -1.7641$ $p_z= 0.0000$
 P' $E= 1.4684$ $p_x= 0.0000$ $p_y= 1.1297$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.7730$ $p_y= 0.6344$ $p_z= 0.0000$

Transverse plane

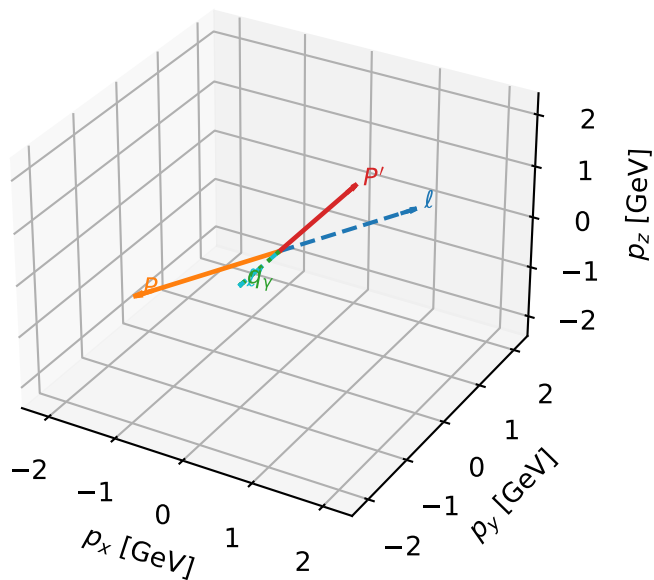


Leading initial-ensemble/final-state components (N=5, retained=98.6%)



Tx auxiliary lepton + Tx proton characteristic max F_3 configuration

3D momenta

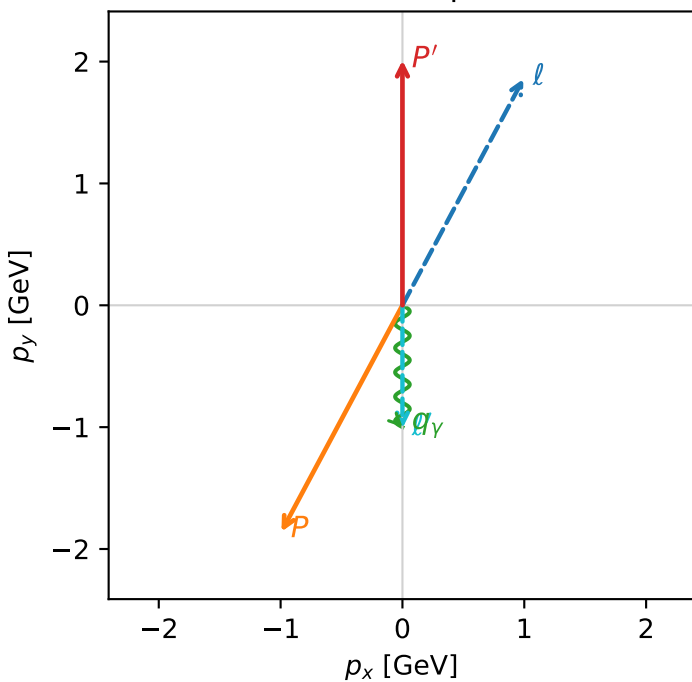


f3_mid_egamma_txtx_region_3 $F_3=0.990669$
 region: mid_Egamma_TxTx_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,l}\rangle \otimes |\bar{T}_{x,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=2.00946$
 $\theta_{in}=1.57079$, $\phi_l=1.07992$, $\phi_p=4.22152$
 $E_\gamma=1$, $\phi_\gamma=4.71239$
 $Q^2=8.37921$, $x_B=0.497775$, $t=-15.937$, $W^2=9.33396$, $y=0.857268$
 $\theta(l', \gamma)=0$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= 0.9932$ $p_y= 1.8582$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -0.9932$ $p_y= -1.8582$ $p_z= 0.0000$
 l' $E= 1.4209$ $p_x= 0.0000$ $p_y= -1.0095$ $p_z= 0.0000$
 P' $E= 2.2176$ $p_x= 0.0000$ $p_y= 2.0095$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.0000$ $p_y= -1.0000$ $p_z= 0.0000$

Transverse plane



$h_e=-1, h_p=+1, h_\gamma=+1$
 auxiliary lepton Tx, proton Tx

$h_e=+1, h_p=-1, h_\gamma=-1$
 auxiliary lepton Tx, proton Tx

Leading initial-ensemble/final-state components (N=2, retained=97.5%)

